

1(a)	X attached to operator ;			1												
1(b)	<table><tr><th>structural gene</th><th>gene product</th><th>role of gene product</th></tr><tr><td><i>lacZ</i></td><td>β-galactosidase</td><td>hydrolyses lactose to form glucose and galactose</td></tr><tr><td><i>lacY</i></td><td>lactose permease or β-galactoside permease</td><td><i>idea</i> that lactose can enter, cell / bacterium</td></tr><tr><td><i>lacA</i></td><td>transacetylase</td><td>transfers an acetyl group from one molecule to another</td></tr></table>	structural gene	gene product	role of gene product	<i>lacZ</i>	β -galactosidase	hydrolyses lactose to form glucose and galactose	<i>lacY</i>	lactose permease or β -galactoside permease	<i>idea</i> that lactose can enter, cell / bacterium	<i>lacA</i>	transacetylase	transfers an acetyl group from one molecule to another			2
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	One mark for each correct column															
1(c)	(enzyme that is only) produced when, the substrate / lactose / inducer, is present ; A in larger amounts			1												

Question	Answer	Marks
1(d)	any four from: B - <i>population remains constant</i> 1 <i>ref. to glucose has been used up ;</i> 2 <i>(so bacterial) replication / division / binary fission, stops / slows ;</i> B/C 3 <i>/lac operon / (named) structural gene(s), switched on / transcribed / expressed</i> or <i>(named) enzyme(s) synthesised ;</i> C – <i>population size increases</i> 4 <i>(so more) lactose enters ;</i> 5 <i>lactose, hydrolysed / broken down, into, glucose / galactose (to make ATP) ;</i> 6 <i>(for bacterial) replication / division / binary fission ;</i>	4